Presentation 2

Links to Power BI Dashboard, Dashboard Code, Presentation:

https://app.powerbi.com/links/KHPs-Jo0ab?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/QueueTimeSummaries_Oct21_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/QueueAnalysis_NumberAnalysis_Oct21_AggregateTeam2.pdf

• What did you do?

This week, our objective was to analyze queue waiting times per menu. To begin, I worked with Jake to define the queue names and the largest queues to analyze. Building off his work in Python, I edited the code to be usable in R which defined the calls that reached queues and were in queues with more than 500 entries (to focus resources on the largest queues). From there we calculated the average and total time in the queue for each contact session ID. To build a further analysis beyond just the counts, I joined the chosen calls back to the full dataset. I then created columns for month name, year, open/closed, and queue wait times in minutes. These will make it easier to extrapolate interpretations of the data from visualizations and filtering.

• How does it help the project?

This analysis told us how resource strain varies across months as well as across menus. While overall the most resources should be allocated to the family and housing menus, there could be rotating staff that handle the increased call volume for education, consumer, and immigration menus throughout the year. For example, education queue average wait duration increases in July with the upcoming school year, having one phone staff member focus on these calls may help reduce queue wait times. This can be applied to April with consumer queue menus and beyond.

• Next steps

My next steps for presentation 3 are to further quantify the traffic and queue wait times per menu. Looking deeply into the submenus and breaking each menu like family and housing into its submenu parts will give a better idea of which queues people are waiting the longest in. This can help with additional resource allocation and better, actionable recommendations.

Presentation 1

Links to Power BI Dashboard, Dashboard code, and Data Import Code:

• https://app.powerbi.com/links/pyRtXmrU-V?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/AggregateDataSummaries_Oct7_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data import/CarDataImport_Oct7_LoganRoever.R_

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/allcallsimport_Oct7_LoganRoever.R

• What did you do?

For the first presentation, I wanted to look at which actions were happening multiple times per call. I started by converting the provided import data code to use in R instead and imported the data from there. I cleaned the data names and decided to calculate a metric which would explore the call activity distribution. For this analysis, I chose to look at calls between 10 - 75 entries because I wanted to avoid sessions that had random selections. The highest number of entries was 262, which we can deduce is from random selection to reach an agent rather than following the menu flow. I also started with 10 entries because I wanted to look at a section of calls that were inefficient, but likely valid attempts, to see where inefficiencies are occurring in the call flow. After identifying the contact session ids that matched this number of entries, I grouped by activity name. From there I calculated the total number of entries for that activity, the number of distinct sessions that selected that activity, and the average times per session the activity was selected.

• How does it help the project?

This should give us an idea of where people are getting "stuck" in an action or where the system is inefficiently directing the call to the same place again. Looking at the activities with more than 1 instance per session should help us identify possible ways to clean up the process and clear confusion / improve efficiency.

• Issues faced (if any)

We were unsure how to combine into one dashboard without the full PowerBI access, so we don't have a single dashboard yet this week. Additionally, there are some entries that I wasn't sure about, such as playmoh300s. When does this occur in a call? Is this just the hold activity?

- Attempts to resolve issues

We discussed our individual plans and got a general understanding of each other's work for this presentation. We chose one person to present all the dashboards so we wouldn't have to switch devices.

- Next steps

My next steps for presentation 2 are to analyze the QueueMenu1 activity and understand which menus have the longest queues. We will try to plot the average waiting time per queue to understand which menus need more attention and resources. The potential issues with this approach using the CAR data is that it may be difficult to identify which menu the customer was in before being sent to the queue.